

Autonomous Vehicle Lane Detection Using OpenCV

Abstract

This project presents a computer vision-based lane detection system for autonomous vehicles. The system processes road video footage and identifies lane markings using image processing techniques. OpenCV was used for video processing, edge detection, and line detection. The detected lanes are highlighted on the original video, providing a visual representation of road boundaries.

Objective

The objective of this project is to develop a lane detection system that can identify road lane markings from dashcam footage using computer vision techniques.

Technologies Used

- Python
- OpenCV
- NumPy
- Visual Studio Code

Methodology

Step 1: Video Acquisition

Road video footage was loaded using OpenCV's VideoCapture function.

Step 2: Grayscale Conversion

The video frames were converted into grayscale to simplify image processing and reduce computational complexity.

Step 3: Gaussian Blur

A Gaussian Blur filter was applied to remove noise and smooth the image.

Step 4: Edge Detection

Canny Edge Detection was used to identify significant edges corresponding to lane boundaries.

Step 5: Hough Transform

Probabilistic Hough Line Transform was applied to detect straight lane lines from the edge-detected image.

Step 6: Lane Overlay

The detected lane lines were drawn and overlaid on the original video frames.

Step 7: Output Generation

The processed frames were saved into an output video file showing detected lanes.

Results

The system successfully detected lane markings and highlighted them on the road video. The output demonstrated the effectiveness of image processing techniques for lane detection applications.

Applications

- Autonomous Vehicles
- Advanced Driver Assistance Systems (ADAS)
- Traffic Monitoring Systems
- Smart Transportation Systems

Future Enhancements

- Region of Interest (ROI) based lane filtering
- Curved lane detection
- Real-time camera integration
- Deep Learning based lane segmentation using U-Net
- Integration with ROS (Robot Operating System)

Conclusion

The project successfully implemented a lane detection system using OpenCV and image processing techniques. The developed solution demonstrates fundamental concepts used in autonomous driving and intelligent transportation systems.